THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 11 & 12: **Electricity, Magnetic Effects of Electric Current** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For a fuse wire, Student A says it should have a low melting point; Student B says it should have a high melting point. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

2. The resistivity of a material depends on: 1

(a) nature of material and temperature **(b)** area
(c) shape **(d)** length

3. If the potential difference across a resistor is doubled, the current becomes: 1

(a) same **(b)** four times **(c)** half **(d)** double

4. If the current through a resistor is doubled, the heat produced becomes: 1

(a) same **(b)** double **(c)** four times **(d)** half

5. The device used to change the resistance in a circuit is: 1

(a) ammeter **(b)** rheostat **(c)** fuse **(d)** voltmeter

6. For a fuse, Student A says it is connected in series; Student B says in parallel. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

7. The neutral wire has insulation of colour: 1

(a) red **(b)** black **(c)** yellow **(d)** green

8. An electric fuse works on the: 1

(a) heating effect of current **(b)** chemical effect
(c) magnetic effect **(d)** photoelectric effect

9. The pattern of field lines inside a solenoid shows that the field is: 1

(a) zero **(b)** the same at all points
(c) strongest at the ends **(d)** circular

10. The current in a straight wire is reversed. The direction of the magnetic field: 1

(a) reverses **(b)** becomes radial **(c)** stays the same **(d)** becomes zero

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is copper used for making electric wires? 2

12. A student says that the resistance of a conductor is independent of its length. Justify or refute. 2

13. Name two ways of increasing the strength of an electromagnet. 2

14. A student says that the earth wire carries current all the time. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. How can three resistors of 2 Ω, 3 Ω and 6 Ω be connected to give a total resistance of 3
(i) 4 Ω
(ii) 1 Ω?

16. An electron enters a magnetic field at right angles. Explain its path and the rule to find the force. 3

17. Why is the series arrangement not used for domestic circuits? (three reasons) 3

OR

The current through a resistor is 0.5 A when 10 V is applied. What will be the current if voltage is increased to 25 V? Find the resistance. 3

18. Explain the role of the earth wire in metal-bodied appliances. 3

OR

How does the magnetic field of a circular loop become uniform at its centre? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Derive $R = \rho l / A$ from the dependence of resistance. 5
(ii) Find the length of a nichrome wire of resistance 10 Ω with area 2×10^{-7} m² ($\rho = 1 \times 10^{-6}$ Ω m).

OR

(i) Explain electric power and its commercial unit. 5
(ii) A 1200 W iron is used 1 h daily and a 60 W bulb 10 h daily. Find the monthly bill (30 days) at ₹5 per unit.

20. (i) Explain the household wiring system with live, neutral and earth wires. 5
(ii) Describe the safety measures used in electric circuits.

OR

(i) Describe the magnetic field of a current-carrying solenoid. 5
(ii) What is an electromagnet?
(iii) How does the polarity of a solenoid depend on the direction of current?

END OF QUESTION PAPER — Total: 40 Marks

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